

Linux Advanced Assignment

1. Describe the content of the top command

Ans : The top command displays the dynamic snapshot of the all the processes running on the machine . It tells the machine up time , the average load the machine takes ,total tasks running on the machine currently , %CPU memory used , space used by user and kernel. It displays the details in a table format like

PID:USER:PR:NI:VIRT:RES:SHR:S:%CPU:%MEM:TIME:COMMAND

PID : It tells about the process id.

USER : It tells about the user .

PR : It shows the priority of the processes running currently.

NI : It shows the priority of the niceness of the processes.

VIRT : It shows the total virtual memory the process has access to.

RES : It shows the actual physical memory (RAM) the process uses .

SHR : It shows the actual shared memory used by process.

%CPU : It displays the actual percentage of resources used by a process of CPU.

%MEM : It does the same as %CPU but only it shows the memory used.

COMMAND : It tells about the command used to invoke the process that is currently being shown.

```
piyush@CK-piyush: ~/Desktop
piyush@CK-piyush:~/Desktop$ top

top - 10:53:39 up 1 day, 1:41, 1 user, load average: 0.25, 0.40, 0.35
Tasks: 285 total, 2 running, 283 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.8 us, 1.7 sy, 0.0 ni, 96.4 id, 0.1 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15696.4 total, 7881.0 free, 4004.1 used, 4409.1 buff/cache
MiB Swap: 62922.0 total, 62922.0 free, 0.0 used, 11692.3 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR   S  %CPU  %MEM     TIME+ COMMAND
 2872 piyush    20   0 4852992 357236 142796 S  12.1  2.2   14:17.69 gnome-shell
   648 root      -51   0     0     0     0 S   7.7  0.0    2:55.42 irq/165-elan_i2c
21664 piyush    20   0 561808  60768  47280 R   4.0  0.4    0:08.73 gnome-terminal-
 5629 piyush    20   0 12.0g 801276 290220 S   2.9  5.0   23:17.46 firefox
 1025 avahi     20   0 10480   6688  4256 S   0.7  0.0    1:27.28 avahi-daemon
18782 piyush    20   0 3177944 555480 129932 S   0.7  3.5   1:39.56 Isolated Web Co
20666 root      0 -20     0     0     0 I   0.7  0.0    0:01.94 kworker/u33:1-i915_flip
22964 piyush    20   0 14536   6120  3824 R   0.7  0.0    0:01.53 top
   18 root      20   0     0     0     0 I   0.4  0.0    0:14.41 rcu_preempt
   556 root     -51   0     0     0     0 S   0.4  0.0    0:12.49 irq/153-iwlwifi:default_queue
 3344 piyush    20   0 543368 12492 10484 S   0.4  0.1    0:03.28 gsd-sharing
 6933 piyush    20   0 3687752 944548 128596 S   0.4  5.9    6:16.43 Isolated Web Co
21051 root      20   0     0     0     0 I   0.4  0.0    0:02.11 kworker/4:3-i915-unordered
22035 root      20   0     0     0     0 I   0.4  0.0    0:00.73 kworker/u32:0-ext4-rsv-conversion
22127 root      20   0     0     0     0 I   0.4  0.0    0:00.22 kworker/1:1-i915-unordered
22755 piyush    20   0 2579936 74892 59204 S   0.4  0.5    0:00.59 Web Content
     1 root      20   0 23584 15000  9620 S   0.0  0.1    0:04.62 systemd
```

2. How to add apt repository in Ubuntu and install new packages.

Ans : To add apt repository , I have used here sudo add-apt-repository command to add an apt repository which takes the ppa archive file to add the repository in ubuntu .

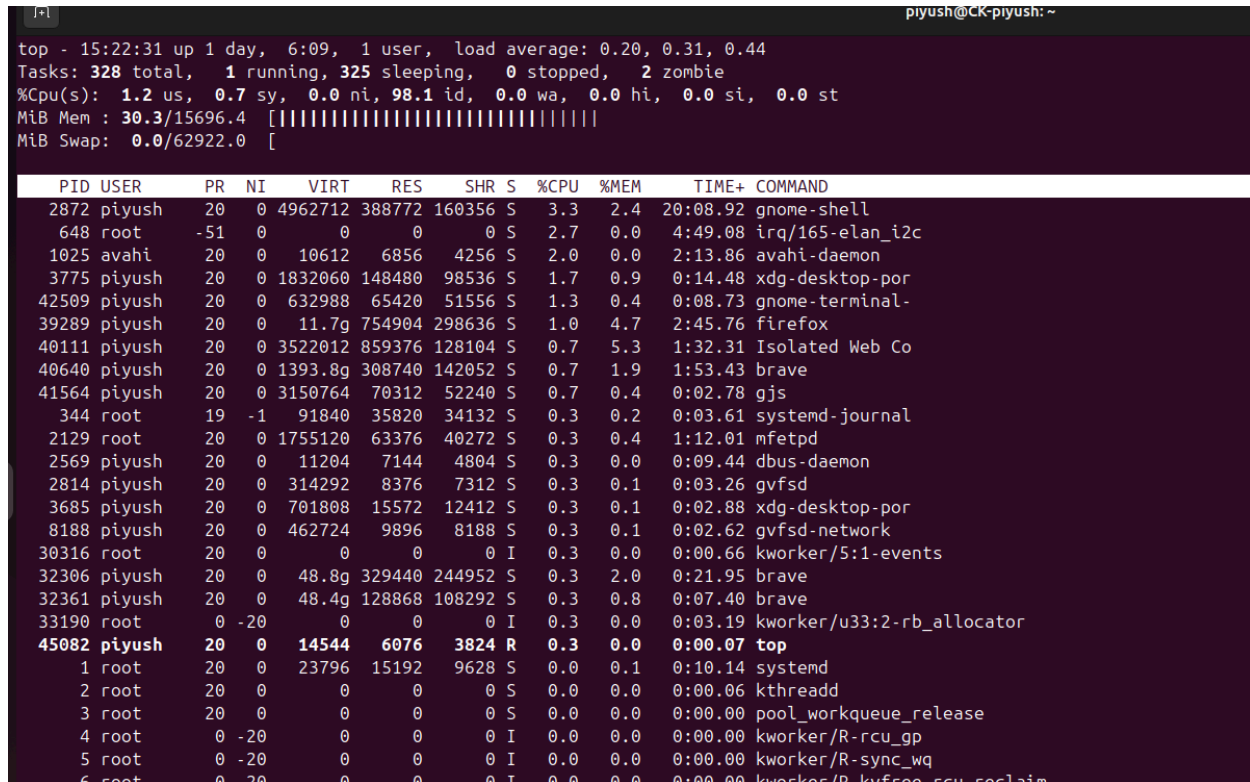
To install the package I can use the sudo apt install command and the packages will be installed.

```
piyush@CK-piyush:~$ sudo add-apt-repository ppa:nginx/stable
[sudo] password for piyush:
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/nginx/stable/ubuntu/
Suites: noble
Components: main
Description:
This PPA contains the latest Stable Release version of the nginx web server software.
**Only Non-End-of-Life Ubuntu Releases are supported in this PPA**
**Development releases of Ubuntu are not officially supported by this PPA, and uploads for those will not be available until actual final releases for those versions**
More info: https://launchpad.net/~nginx+archive/ubuntu/stable
Adding repository.
Press [ENTER] to continue or Ctrl-c to cancel.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Ign:3 https://ppa.launchpadcontent.net/nginx/stable/ubuntu noble InRelease
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Err:5 https://ppa.launchpadcontent.net/nginx/stable/ubuntu noble Release
      404 Not Found [IP: 185.125.190.80 443]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.6 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,284 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Reading package lists... Done
E: The repository 'https://ppa.launchpadcontent.net/nginx/stable/ubuntu noble Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
piyush@CK-piyush:~$
```

```
piyush@CK-piyush:~$ sudo apt install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libgl1-amd-dri libglapi-mesa libllvm19
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 15 not upgraded.
Need to get 564 kB of archives.
After this operation, 1,596 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 564 kB in 5s (124 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 215458 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.5) ...
* Upgrading binary nginx
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
```

3. List all the processes in the descending order of memory and CPU consumed by them.

Ans : The command I use is the top command because it makes the job easier as by default it list the processes in descending order first by CPU consumed and for listing the processes in descending order of the memory consumed I just need to press M and it will do the job for me .



```
top - 15:22:31 up 1 day, 6:09, 1 user, load average: 0.20, 0.31, 0.44
Tasks: 328 total, 1 running, 325 sleeping, 0 stopped, 2 zombie
%Cpu(s): 1.2 us, 0.7 sy, 0.0 ni, 98.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 30.3/15696.4 [|||||||||||||||||||||||||]
MiB Swap : 0.0/62922.0 [ ]

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
 2872 piyush    20   0 4962712 388772 160356 S   3.3   2.4   20:08.92 gnome-shell
   648 root      -51   0     0     0     0  S   2.7   0.0    4:49.08 irq/165-elan_i2c
 1025 avahi     20   0  10612   6856  4256  S   2.0   0.0    2:13.86 avahi-daemon
 3775 piyush    20   0 1832060 148480 98536  S   1.7   0.9    0:14.48 xdg-desktop-por
 42509 piyush    20   0  632988  65420 51556  S   1.3   0.4    0:08.73 gnome-terminal-
 39289 piyush    20   0   11.7g 754904 298636 S   1.0   4.7    2:45.76 firefox
 40111 piyush    20   0 3522012 859376 128104 S   0.7   5.3    1:32.31 Isolated Web Co
 40640 piyush    20   0 1393.8g 308740 142052 S   0.7   1.9    1:53.43 brave
 41564 piyush    20   0 3150764 70312 52240  S   0.7   0.4    0:02.78 gjs
   344 root      19  -1   91840  35820 34132  S   0.3   0.2    0:03.61 systemd-journal
 2129 root      20   0 1755120 63376 40272  S   0.3   0.4    1:12.01 mftpd
 2569 piyush    20   0   11204   7144  4804  S   0.3   0.0    0:09.44 dbus-daemon
 2814 piyush    20   0   314292  8376  7312  S   0.3   0.1    0:03.26 gvfsd
 3685 piyush    20   0   701808 15572 12412  S   0.3   0.1    0:02.88 xdg-desktop-por
 8188 piyush    20   0 462724   9896  8188  S   0.3   0.1    0:02.62 gvfsd-network
 30316 root      20   0     0     0     0  I   0.3   0.0    0:00.66 kworker/5:1-events
 32306 piyush    20   0   48.8g 329440 244952 S   0.3   2.0    0:21.95 brave
 32361 piyush    20   0   48.4g 128868 108292 S   0.3   0.8    0:07.40 brave
 33190 root      0 -20     0     0     0  I   0.3   0.0    0:03.19 kworker/u33:2-rb_allocator
45082 piyush    20   0  14544   6076  3824  R   0.3   0.0    0:00.07 top
    1 root      20   0   23796 15192  9628  S   0.0   0.1    0:10.14 systemd
    2 root      20   0     0     0     0  S   0.0   0.0    0:00.06 kthreadd
    3 root      20   0     0     0     0  S   0.0   0.0    0:00.00 pool_workqueue_release
    4 root      0 -20     0     0     0  I   0.0   0.0    0:00.00 kworker/R-rcu_gp
    5 root      0 -20     0     0     0  I   0.0   0.0    0:00.00 kworker/R-sync_wq
    6 root      0 -20     0     0     0  I   0.0   0.0    0:00.00 kworker/R-kvfree_rcu_reclaim
```

4. In which file you can write commands which you want to run whenever Linux system starts/restarts?

Ans : To write commands which can be used run whenever Linux system starts/restarts the folder used is /etc/systemd/system/myservice.service . This file create the service to be executed whenever the system starts or restarts.

The service file should be created first to create the service .

Using commands sudo systemctl enable myservice.service

Sudo systemctl start myservice.service command we can enable and start the service.

All the services are present inside the etc/systemd/system directory.

```
piyush@CK-piyush: /etc/systemd/system
piyush@CK-piyush:~$ cat /etc/systemd
cat: /etc/systemd: Is a directory
piyush@CK-piyush:~$ cd /etc/systemd
piyush@CK-piyush:/etc/systemd$ cd system
piyush@CK-piyush:/etc/systemd/system$ ls
bluetooth.target.wants
cloud-config.target.wants
cloud-final.service.wants
cloud-init.target.wants
dbus-fi.wi.wpa_supplicant1.service
dbus-org.bluez.service
dbus-org.freedesktop.Avahi.service
dbus-org.freedesktop.ModemManager1.service
dbus-org.freedesktop.nm-dispatcher.service
dbus-org.freedesktop.oom1.service
dbus-org.freedesktop.resolve1.service
dbus-org.freedesktop.thermal.service
dbus-org.freedesktop.timesync1.service
display-manager.service
display-manager.service.wants
emergency.target.wants
final.target.wants
getty.target.wants
graphical.target.wants
hibernate.target.wants
hybrid-sleep.target.wants
ma.service
mcafee.ma.service
multi-user.target.wants
network-online.target.wants
oem-config.service.wants
```

5. Restart cron service.

Ans : The command I used to restart the cron service for my Ubuntu distribution is using systemctl because it does not support init . The exact command used is
sudo systemctl restart cron

By using sudo systemctl status I can see the active cron jobs and then by running again I can see that my cron service restarted again .

```
piyush@CK-piyush:~$ sudo systemctl status cron
● cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-22 16:45:48 IST; 26min ago
     Docs: man:cron(8)
    Main PID: 8726 (cron)
      Tasks: 1 (limit: 18628)
    Memory: 532.0K (peak: 6.5M)
       CPU: 69ms
    CGroup: /system.slice/cron.service
            └─8726 /usr/sbin/cron -f -P

Jan 22 16:45:48 CK-piyush cron[8726]: (CRON) INFO (Skipping @reboot jobs -- not system startup)
Jan 22 16:55:01 CK-piyush CRON[9056]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jan 22 16:55:01 CK-piyush CRON[9057]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Jan 22 16:55:01 CK-piyush CRON[9056]: pam_unix(cron:session): session closed for user root
Jan 22 17:00:01 CK-piyush CRON[9377]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jan 22 17:00:01 CK-piyush CRON[9378]: (root) CMD (/opt/McAfee/agent/scripts/ma checkhealth >/dev/null 2>/dev/null)
Jan 22 17:00:01 CK-piyush CRON[9377]: pam_unix(cron:session): session closed for user root
Jan 22 17:05:01 CK-piyush CRON[9593]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jan 22 17:05:01 CK-piyush CRON[9594]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Jan 22 17:05:01 CK-piyush CRON[9593]: pam_unix(cron:session): session closed for user root
piyush@CK-piyush:~$ sudo systemctl restart cron
piyush@CK-piyush:~$ sudo crontab -l
no crontab for root
piyush@CK-piyush:~$ sudo systemctl status cron
● cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-22 17:12:37 IST; 55s ago
     Docs: man:cron(8)
    Main PID: 9729 (cron)
      Tasks: 1 (limit: 18628)
    Memory: 340.0K (peak: 620.0K)
       CPU: 1ms
    CGroup: /system.slice/cron.service
            └─9729 /usr/sbin/cron -f -P

Jan 22 17:12:37 CK-piyush systemd[1]: Started cron.service - Regular background program processing daemon.
Jan 22 17:12:37 CK-piyush cron[9729]: (CRON) INFO (pidfile fd = 3)
Jan 22 17:12:37 CK-piyush cron[9729]: (CRON) INFO (Skipping @reboot jobs -- not system startup)
piyush@CK-piyush:~$
```

6. Create an ext4 filesystem.

Ans : To create the ext4 filesystem , first I use lsblk to see the list of all block storage and partitions on the system. Then using the command `sudo mkfs.ext4 -L data1 /dev/nvme0n1p4` to create the ext4 filesystem, Then we can run the lsblk command again to see if the filesystem is created.

```

piyush@CK-piyush:~$ lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0                 7:0      0    4K  1 loop /snap/bare/5
loop1                 7:1      0 187.9M  1 loop /snap/brave/586
loop2                 7:2      0   74M  1 loop /snap/core22/2193
loop3                 7:3      0  73.9M  1 loop /snap/core22/2216
loop4                 7:4      0  258M  1 loop /snap/firefox/5751
loop5                 7:5      0 251.6M  1 loop /snap/firefox/7672
loop6                 7:6      0  11.1M  1 loop /snap/firmware-updater/167
loop7                 7:7      0  18.5M  1 loop /snap/firmware-updater/210
loop8                 7:8      0   516M  1 loop /snap/gnome-42-2204/202
loop9                 7:9      0 531.4M  1 loop /snap/gnome-42-2204/247
loop10                7:10     0  91.7M  1 loop /snap/gtk-common-themes/1535
loop11                7:11     0  10.8M  1 loop /snap/snap-store/1248
loop12                7:12     0  48.1M  1 loop /snap/snapd/25935
loop13                7:13     0  44.4M  1 loop /snap/snapd/23545
loop14                7:14     0   576K  1 loop /snap/snapd-desktop-integration/315
loop15                7:15     0   568K  1 loop /snap/snapd-desktop-integration/253
loop16                7:16     0 210.4M  1 loop /snap/thunderbird/644
loop17                7:17     0 226.6M  1 loop /snap/thunderbird/933
loop18                7:18     0  63.8M  1 loop /snap/core20/2686
loop19                7:19     0  128M  1 loop /snap/obsidian/55
nvme0n1              259:0     0 476.9G  0 disk
├─nvme0n1p1           259:1     0 139.7G  0 part /
├─nvme0n1p2           259:2     0    1G  0 part /boot/efi
├─nvme0n1p3           259:3     0 181.6G  0 part /home
├─nvme0n1p4           259:4     0  93.1G  0 part
└─nvme0n1p5           259:5     0  61.4G  0 part [SWAP]
piyush@CK-piyush:~$ # sudo mkfs.ext4 -L data1 /nvme0n1p4

```

7. Mount the created filesystem on /partition directory.

Ans : So first I created the /partition directory using the simple mkdir command then Using sudo mount -L data1 /partition I mounted the created filesystem on the partition directory .

To verify if the partition directory has been mounted successfully I ran the command df -h to see if filesystem is mounted successfully.

```

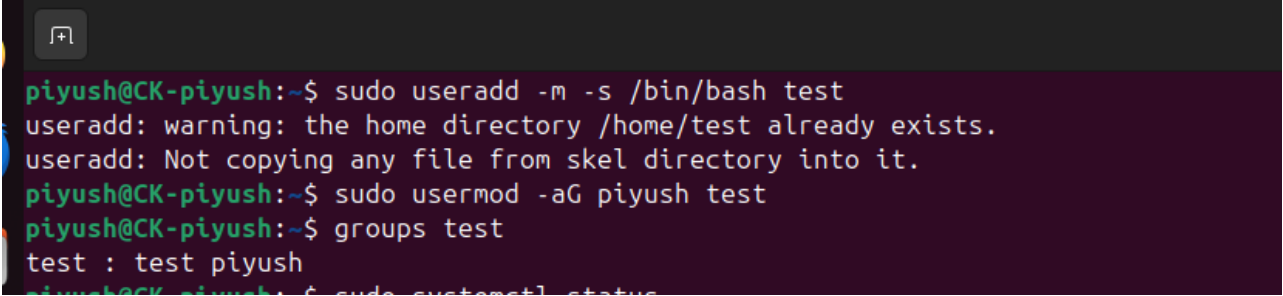
piyush@CK-piyush:~$ # sudo mkdir partition
piyush@CK-piyush:~$ # sudo mount -L data1 partition
piyush@CK-piyush:~$ # df -h
piyush@CK-piyush:~$ df -h

```

8. Create a user and add it to one secondary group.

Ans : To create a new user I use the command `sudo useradd` , this create new user for me . Next by using `usermod` command with flags `-aG` I add the newly created user to the secondary group which is my current user i.e. `piyush`.

By using the `groups` command I can check that if the test user is added successfully as the secondary group.



```
piyush@CK-piyush:~$ sudo useradd -m -s /bin/bash test
useradd: warning: the home directory /home/test already exists.
useradd: Not copying any file from skel directory into it.
piyush@CK-piyush:~$ sudo usermod -aG piyush test
piyush@CK-piyush:~$ groups test
test : test piyush
piyush@CK-piyush:~$ sudo systemctl status
```

9. Give this user full access (without password).

Ans : First step would be to open the sudo /sbin/visudo file , this will open the editor to edit the sudoers file , in the end I added test ALL = (ALL) NOPASSWD:ALL , which basically give this test user full access to the system without using the password.

```
GNU nano 7.2 /etc/sudoers.tmp

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d

test ALL=(ALL) NOPASSWD: ALL
```

```
test@CK-piyush: /home/piyush

piyush@CK-piyush:~$ sudo /sbin/visudo
[sudo] password for piyush:
Sorry, try again.
[sudo] password for piyush:
visudo: /etc/sudoers.tmp unchanged
piyush@CK-piyush:~$ sudo su test
test@CK-piyush:/home/piyush$ whoami
test
test@CK-piyush:/home/piyush$
```

10. Delete the created user after taking backup of it home directory.

Ans : First I would compress the home directory in test user using the tar command and create a compressed directory version to create a backup.

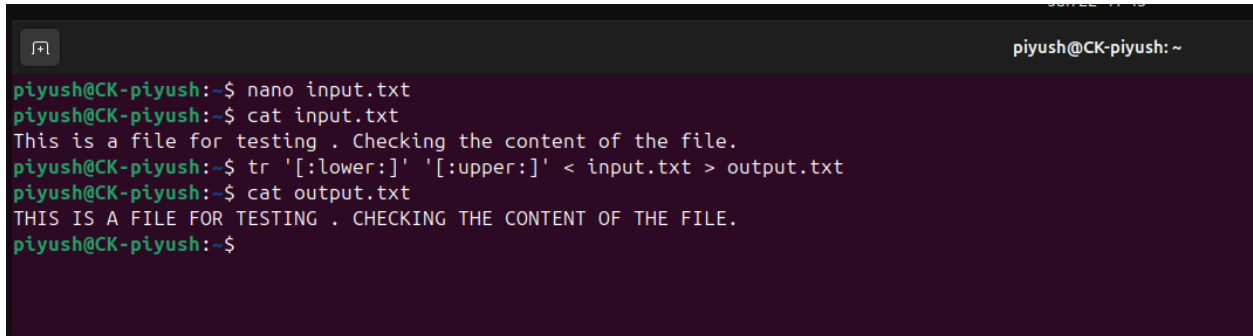
The I would use the command sudo userdel test , this would delete the user for me and I would have the backup . I could have used cp command to store the backup but that would not be as efficient as tar compressed file , because some files may have been left due to some security reasons .

```
piyush@CK-piyush: ~  
piyush@CK-piyush:~$ sudo tar czvf /root/test_backup.tar /home/test  
[sudo] password for piyush:  
tar: Removing leading '/' from member names  
/home/test/  
/home/test/.cache/  
/home/test/.cache/motd.legal-displayed  
/home/test/.local/  
/home/test/.local/share/  
/home/test/.local/share/nano/  
/home/test/.local/state/  
/home/test/.local/state/wireplumber/  
/home/test/.local/state/wireplumber/restore-stream  
/home/test/etc/  
/home/test/etc/passwd_backup  
/home/test/.bash_logout  
/home/test/.bash_history  
/home/test/.profile  
/home/test/tar_files.tar.gz  
/home/test/.bashrc  
/home/test/output  
/home/test/snap/  
/home/test/snap/snapd-desktop-integration/  
/home/test/snap/snapd-desktop-integration/common/  
/home/test/snap/snapd-desktop-integration/common/.cache/  
/home/test/snap/snapd-desktop-integration/common/.cache/gdk-pixbuf-loaders.cache  
/home/test/snap/snapd-desktop-integration/common/.cache/desktop-runtime-date  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/immodules.cache  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-viqr.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-wayland.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-am-et.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-xim.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-cedilla.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-ibus.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-thai.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-fcitx.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-waylandgtk.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-inuktitut.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-ti-er.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-broadway.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-ti-et.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-cyrillic-translit.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-ipa.so  
/home/test/snap/snapd-desktop-integration/common/.cache/immodules/in-multipress.so  
/home/test/snap/snapd-desktop-integration/common/.cache/gio-modules/
```

```
piyush@CK-piyush:~$ sudo userdel test  
piyush@CK-piyush:~$ id test  
id: 'test': no such user  
piyush@CK-piyush:~$
```

11. Create a file with some content. Change all lower case letter to upper case letter and save output to another file using redirections.

Ans : To create a file I used nano input.txt command which opens the editor for me and I added the content accordingly. Then I use tr i.e translate to change the lower case letter to uppercase letter present inside the file. Using '[:lower:]' '[:upper:]' and redirection operation by < taking the input and then redirecting it > to output.txt.

A terminal window with a dark purple background. The prompt is 'piyush@CK-piyush: ~'. The user enters 'nano input.txt', then 'cat input.txt', which outputs 'This is a file for testing . Checking the content of the file.'. Then the user enters 'tr '[:lower:]' '[:upper:]' < input.txt > output.txt', followed by 'cat output.txt', which outputs 'THIS IS A FILE FOR TESTING . CHECKING THE CONTENT OF THE FILE.'. The prompt returns to 'piyush@CK-piyush:~\$'.

```
piyush@CK-piyush:~$ nano input.txt
piyush@CK-piyush:~$ cat input.txt
This is a file for testing . Checking the content of the file.
piyush@CK-piyush:~$ tr '[:lower:]' '[:upper:]' < input.txt > output.txt
piyush@CK-piyush:~$ cat output.txt
THIS IS A FILE FOR TESTING . CHECKING THE CONTENT OF THE FILE.
piyush@CK-piyush:~$
```

12. Set nice value of a process to -1.

Ans : The command I used to set the process's nice value to -1 is `sudo nice -n -1 top`, it does open the dynamic view of processes running with a new process created with nice value of -1.

I tried `nice -n -1 top`, but permission was denied because to set the nice property of a process sudo privileges are required.

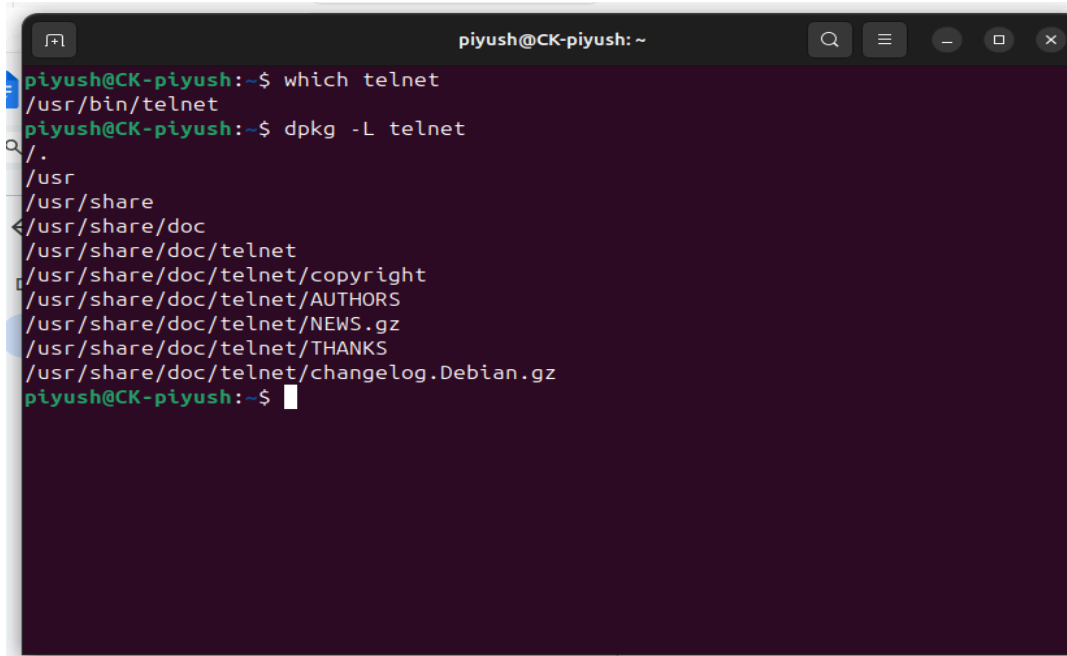
```
piyush@CK-piyush: ~  
piyush@CK-piyush:~$ sudo nice -n -1 top  
top - 21:09:09 up 4:58, 1 user, load average: 0.34, 0.67, 0.56  
Tasks: 317 total, 2 running, 314 sleeping, 0 stopped, 1 zombie  
%Cpu(s): 2.3 us, 2.1 sy, 0.0 ni, 95.6 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 15696.4 total, 7209.7 free, 4830.5 used, 4537.7 buff/cache  
MiB Swap: 62922.0 total, 62922.0 free, 0.0 used, 10865.9 avail Mem  


| PID   | USER     | PR  | NI  | VIRT    | RES    | SHR    | S | %CPU | %MEM | TIME+   | COMMAND                 |
|-------|----------|-----|-----|---------|--------|--------|---|------|------|---------|-------------------------|
| 2598  | piyush   | 20  | 0   | 5029408 | 342212 | 141996 | S | 12.0 | 2.1  | 5:56.97 | gnome-shell             |
| 12069 | piyush   | 20  | 0   | 629556  | 62104  | 50492  | R | 8.4  | 0.4  | 0:03.53 | gnome-terminal-         |
| 899   | root     | -51 | 0   | 0       | 0      | 0      | S | 7.2  | 0.0  | 1:40.61 | irq/165-elan_i2c        |
| 18    | root     | 20  | 0   | 0       | 0      | 0      | I | 1.2  | 0.0  | 0:06.20 | rcu_preempt             |
| 814   | systemd+ | 20  | 0   | 17560   | 7752   | 6836   | S | 1.2  | 0.0  | 0:01.87 | systemd-oomd            |
| 3301  | piyush   | 20  | 0   | 12.1g   | 858828 | 346412 | S | 1.2  | 5.3  | 8:50.82 | firefox                 |
| 4296  | piyush   | 20  | 0   | 3140992 | 480368 | 135788 | S | 1.2  | 3.0  | 3:06.15 | Isolated Web Co         |
| 12428 | root     | 0   | -20 | 0       | 0      | 0      | I | 1.2  | 0.0  | 0:01.33 | kworker/u33:2-i915_flip |
| 13993 | root     | 20  | 0   | 0       | 0      | 0      | I | 1.2  | 0.0  | 0:00.05 | kworker/u32:3-i915      |
| 14025 | root     | 19  | -1  | 14536   | 6116   | 3824   | R | 1.2  | 0.0  | 0:00.05 | top                     |
| 1     | root     | 20  | 0   | 23608   | 14980  | 9584   | S | 0.0  | 0.1  | 0:03.04 | systemd                 |


```

13. Get list of all files used by “telnet”.

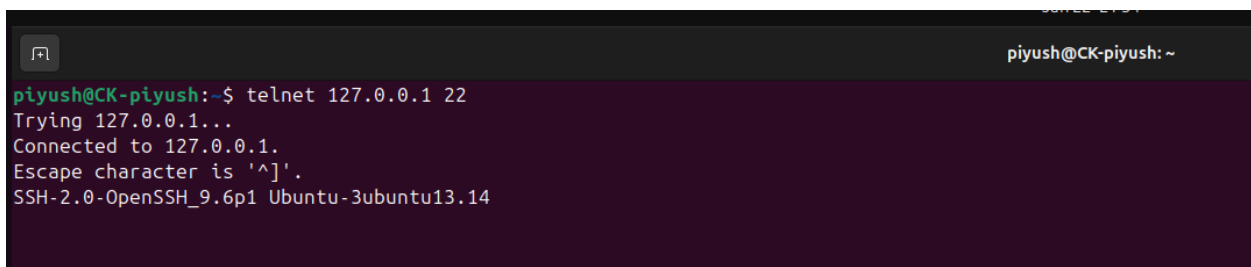
Ans : To locate the binary executable file we can use which followed by the command name and it will show the directory where the telnet binary is stored. Then by using the package manager we can see the files used by the telnet command . Using dpkg -L we can see all the files used by telnet



```
piyush@CK-piyush: ~  
piyush@CK-piyush:~$ which telnet  
/usr/bin/telnet  
piyush@CK-piyush:~$ dpkg -L telnet  
./.  
/usr  
/usr/share  
/usr/share/doc  
/usr/share/doc/telnet  
/usr/share/doc/telnet/copyright  
/usr/share/doc/telnet/AUTHORS  
/usr/share/doc/telnet/NEWS.gz  
/usr/share/doc/telnet/THANKS  
/usr/share/doc/telnet/changelog.Debian.gz  
piyush@CK-piyush:~$
```

14. Check if port 22 is listening using netstat and telnet command.

Ans : To check if port 22 is listening using telnet the command I used is simply telnet 127.0.0.1 22 , this will show if port 22 is listening.
Now using telnet command , I use the flags -ntl which will display the ports which are listening in numerical format -t flag will display tcp ports as well .



```
piyush@CK-piyush: ~  
piyush@CK-piyush:~$ telnet 127.0.0.1 22  
Trying 127.0.0.1...  
Connected to 127.0.0.1.  
Escape character is '^]'.  
SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
```

connection closed by foreign host.

piyush@CK-piyush:~\$ netstat -ntl

Active Internet connections (only servers)

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	127.0.0.53:53	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:631	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.54:53	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:80	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:22	0.0.0.0:*	LISTEN
tcp6	0	0	:::8081	:::*	LISTEN
tcp6	0	0	:::1:631	:::*	LISTEN
tcp6	0	0	:::80	:::*	LISTEN
tcp6	0	0	:::22	:::*	LISTEN

piyush@CK-piyush:~\$

15. Create a cron job which runs once in a week at 23:45.

Ans : To create a new cron job that run once in a week at 23.45 , I first opened the editor using `crontab -e` , that opened a editor , I wrote a simple cron job that will simply echo the message “This is a cron job running once a week at 23.45” .

Now to verify if the cronjob is created , I used `crontab -l` that lists the active cronjobs .

```
piyush@CK-piyush:~$ crontab -e
no crontab for piyush - using an empty one

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano      <---- easiest
 2. /usr/bin/vim.tiny
 3. /bin/ed

Choose 1-3 [1]: 1
crontab: installing new crontab
piyush@CK-piyush:~$ crontab -e
No modification made
piyush@CK-piyush:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
45 23 * * 0 echo "This is a cron job running oncea week at 11.45"
piyush@CK-piyush:~$
```



```
piyush@CK-piyush:~$ crontab -e
no crontab for piyush - using an empty one

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano      <---- easiest
 2. /usr/bin/vim.tiny
 3. /bin/ed

Choose 1-3 [1]: 1
crontab: installing new crontab
piyush@CK-piyush:~$ crontab -e
No modification made
piyush@CK-piyush:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
45 23 * * 0 echo "This is a cron job running once a week at 11.45"
piyush@CK-piyush:~$
```